

ARTIFICIAL INTELLIGENCE FOR TELECOMMUNICATION CRYPTOGRAPHY

TCP/IP Network Administration- EETS 7302

Group No. 2:

VISHAL KHOT
VISHAKHA KAMBLE
MIHEER BHILARE
ABEER ALBADANI

Under the guidance of Professor

Dr. Hakki Cankaya

Master's in Telecommunication and Network Engineering
Southern Methodist University

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Roles accomplished by Group Members:

VISHAL KHOT (LEADER) - RESEARCH/ DEMONSTRATION/ REPORT

VISHAKHA KAMBLE - RESEARCH/ DEMONSTRATION/ REPORT

MIHEER BHILARE - DEMONSTRATION/ CODING/ REPORT

ABEER ALBADANI - RESEARCH/ REPORT

1. INTRODUCTION

1.1 What is AI...?

AI has the ability of Human intelligence which is processed by computer systems. The main aspect of AI is Machines that have Human Intelligence. In simpler words Machines are artificially incorporated with human like intelligence to perform the tasks a normal human does. Artificial Intelligence is also used in our day-to-day lifestyle like Virtual Smart assistants like Siri Alexa Google Cortana. Ai helps in all firms like education, healthcare systems, smart homes, Automobile's, cybersecurity, Robotics, Banking, Entertainment, Smart phones, social media, video games it is used in every sector. This Leads us to the exploration of how AI technology and applications are being used in the Networking domains and how will it help to us all to ease and grow the telecommunication world. A large amount of Artificial intelligence applications in network industries have the capability to minimize Network Utilization costs and establishing new network costs, boosting performance, strengthening customer service and assist the development and introduction of fresh services. This research discovers and inspect the scale of AI applications which are so far in place and those anticipated to be in place soon. At this point, however as AI becomes increasingly integrated into Network operations and help many sectors of world like primarily healthcare, banking, Retail and Ecommerce. In future we can see bright things coming with the use of AI in Networking field.

1.2 Why did we choose AI...?

We have selected the topic of Artificial Intelligence for Telecommunication. The reason for selecting this topic was very clear as the telecom sector is rapidly emerging new technologies and showing advancement innovation, growth in distinct fields of AI. As everyone being from a Telecom background it is quite essential to stay proactive with the ongoing trends in the industry. As our generation is moving toward the advanced information age, the key element for this change would be the rapid and profound transformations faced by the Telecommunications Network Field. It's been said that to achieve such alteration systematically Artificial Intelligence Technology is required. Our motivation behind this research is even though for those Engineers and especially Network administrators who are already well experienced with telecommunication but may be unsure about Artificial Intelligence Technology. The relationship between two rapidly progressing technologies Artificial Intelligence and Telecommunications is Examined. It is found that both technologies share familiar drivers, which contribute us to the mapping between future telecommunications trends and AI capabilities.

2. INDUSTRIAL USES OF AI:

- AI for NETWORK OPTIMIZATION
- AI for PREDICTIVE MAINTENANANCE
- AI for VIRTUAL ASSISTANCE
- AI for CRYPTOGRAPHY

AI for NETWORK OPTIMIZATION:

AI helps the Customer Service Provider (CSP) to build the self-optimizing networks. Due to this the operators have the ability to directly optimize the network quality which is based on the information of traffic by the region and the time zone. The network automation and the intelligence will help to enable better root cause analysis and prediction of issues.

Let's consider a real-life example by AT&T as they have come up to an innovative solution like they have used AI to support its maintenance procedures. The company is testing a drone as to expand its LTE network coverage and to utilize the analysis of the video data which is captured by the drones for the technical support and the maintenance of all its cell tower.

AI for PREDICTIVE MAINTENANCE:

AI predictive analysis helps the telecom industry to provide better services by the following functions like utilizing the data, algorithms, and machine learning techniques to predict future results based on the historical data. This means that the operators are using the data driven insights as to monitor the state of the equipment and assume failure based on the patterns and depending upon that fix problems with communication hardware for example cell tower, power lines, data center servers and the set top box at the customers' homes.

AI for VIRTUAL ASSISTANCE:

A user that completes tasks by understanding natural language voice commands. They are basically cloud based programs and they require internet connection of the devices. Types of virtual assistant chatbots are Alexa, Siri, Google Assistant, Cortana. According to research done by JUNIPER they have observed that people have learned to automate and scale one on one conversations so perfectly that they are projected to cut the expenses of the business up to 8 billion annually in 2022.

3. Why we chose cryptography?

From the start, cryptography, and machine learning both were closely correlated. This is because they share a regular target. To find the right key for decryption is one of the main targets of Cryptanalyst's, on the other hand to find a perfect solution in a bigger space of achievable solutions is the target of machine learning's. Cryptography played a massive role in the direction of World War II, also few of the first working computers were committed to cryptanalytic tasks.

3.1 CRYPTOGRAPHY

Cryptography is a process of transforming readable plain text messages into an unintelligible form and later the reversible of that process. Cryptography can be used to send sensitive and confidential information or data over an untrusted network such as the internet and it uses authentication and encryption method to do that. Cryptography provides some following services to data: 1) Authenticity which is a proof of source like proving that the data really come from the legitimate source. 2) confidentiality which is privacy and secrecy which means the user can send data over an untrusted network and if someone is able to sniff that traffic, they are not able to read it. 3) integrity meaning that the data is not changed in transit and finally 4) non-repudiation or non- deniability, which is quite similar to authenticity. With cryptography we got both symmetric and asymmetric encryption.

SYMMETRIC ENCRYPTION

In symmetric encryption, the admin uses the same shared key for both encryption and decryption of the data. We can relate this shared key to a password. This shared key must be known by both the receiver and the sender and must be kept a secret. Symmetric encryption is fast and therefore it used for large transmission such as email, secure web traffic, HTTPS and IPsec for VPN tunnels. Algorithms that are used for symmetric encryption includes DES, 3DES, AES, SEAL. DES and 3DES are really older algorithm and it is not used so commonly now. The most commonly used algorithm in real world is AES.

WORKING OF SYMMETRIC ENCRYPTION:

The below figure shows the working of Symmetric Encryption and how it provides confidentiality.

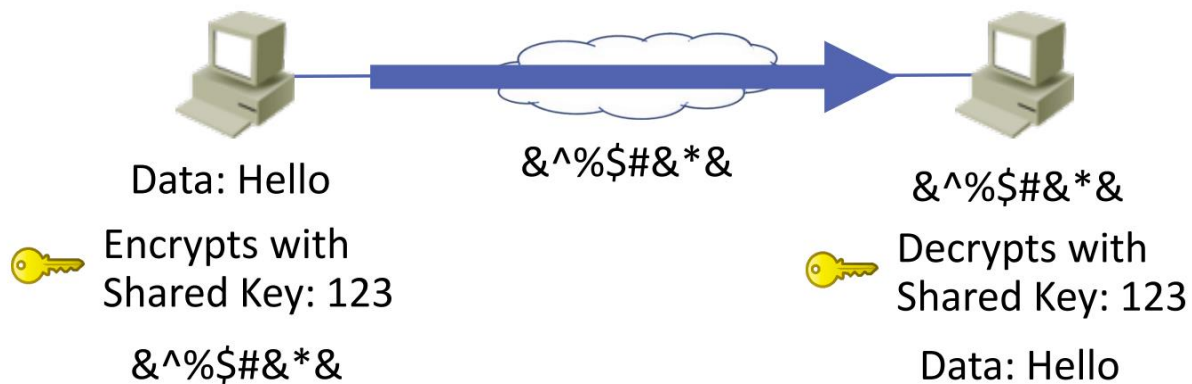


Figure 1

The host(left) and he have a data named Hello which the host want to send securely over the internet to another host(right). The sender encrypts the HELLO message with a shared key called 123 and when it does that, the plain message becomes some garble encrypted data as shown in fig1. This encrypted data is sent across the untrusted network. So, if there is a sniffer on this untrusted network, he can sniff the data but he cannot read the actual data what the sender is sending, instead he will see the encrypted garble copy of the actual data. When the encrypted data comes to the intended target, the receiver decrypt the garble message with the same shared key 123 and the message comes out into the readable format which in this case is HELLO.

ASYMMETRIC ENCRYPTION

Asymmetric encryption uses private and public key pairs. The way it works is that the data encrypted with the private key is only decrypted by the public key and the message encrypted with the public key can only be decrypted by private key. Data encrypted with public key cannot be decrypted with the public key. Only the private key must be kept a secret and with asymmetric encryption the public key can be shared with everyone. The public key can be known in the public domain. The asymmetric encryption is slow compared to symmetric encryption and because of that this method is used for smaller transmission like digital signatures. Algorithms for asymmetric encryption includes RSA and ECDSA.

WORKING OF ASYMMERIC ENCRYPTION

The fig below shows the working of Asymmetric Encryption.

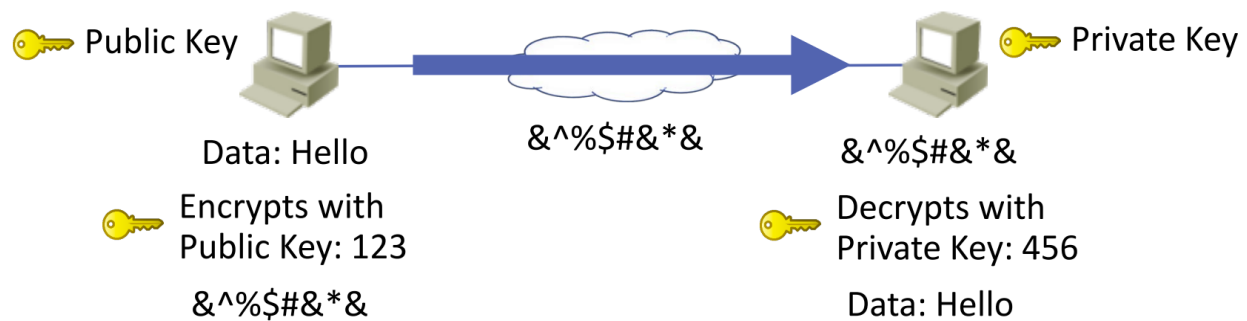


Figure 2

The sender on the left has a public key and the receiver on the right has the matching private key. The sender has the public key for the host on the right and wants to send the data HELLO over the unsecured network. It takes out data of HELLO and encrypts it with public key and then makes the plain text into a cipher text. The host on the left does not know the private key of the host on the right. It only knows the public key of its own. Then the data is send over the unsecured network(internet) to the destination target. The destination host receives the garble data and decrypts it with their own private key which brings out the original plain text HELLO. With this it only allows anybody to send data securely to the host with the private key because all the hosts are allowed to have the public key. The destination host is the only host which has the private key and nobody else has that, so this is the only one who can read the message that have been encrypted with its public key. Other host with the public key cannot read the messages.

3.2 How is AI changing Cryptography-- algorithms explanation (AES)

- Uses Advanced Algorithm's to look for patterns within the data.
- The integration of artificial intelligence technology with the telecommunications industry is aiding the various communication service providers in managing, optimizing, and also maintaining the infrastructures. Also, in managing the customer support functionalities.
- Optimizing Mobile Tower Operations
- Better Customer Service

AES algorithms overview.

AES is 128-bit block cipher that means it takes 128 bits of message and it encrypts it into 128 bits of ciphertext with some key now that key can be either 128, 192 or 256 bits and that gives you just varying amounts of security. These bits are specified as the part of the AES standard.

AES likes to arrange things in a grid a 4x4 grid like a 428 bit, so we have our message which is 128 bit that's 16 bytes as a 4x4 grid. So bytes will be arranged in a 4x4 grid as show in the picture below.

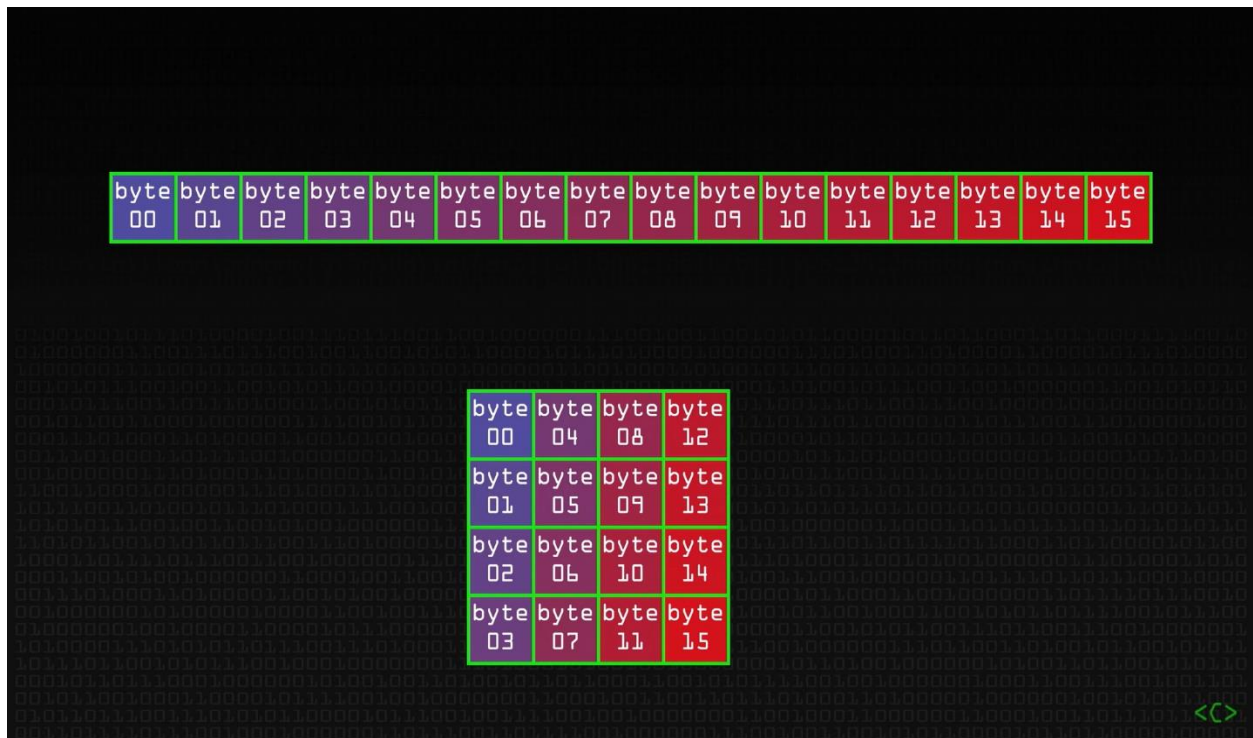


Figure 3

So, what we are doing is we are taking our 128-bit message and we are just laying it out in this order. Then we will start the process by permute, by substitute byte and then we can transform it into some way where an attacker can't read what the message used to be. But remember one thing everything in AES happens on this 4x4 grid.

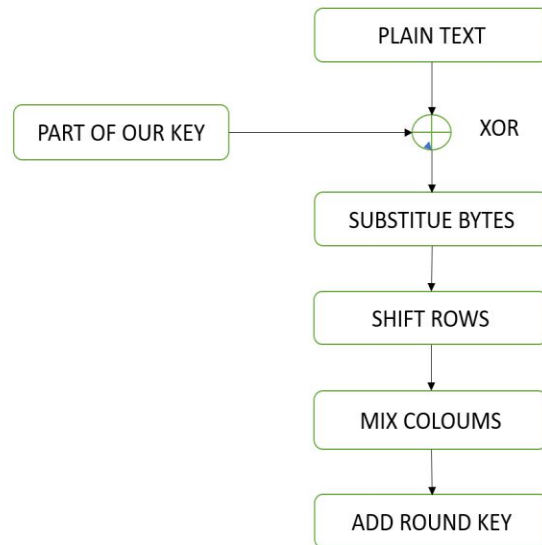


Figure 4

So, referring to the above flow chart, this is our plain text and we are going to first bring in a part of our key and we are going to perform an XOR operation, then we are going to do our rounds so that is going to be substitute bytes then we are going to shift rows and then finally we are going to mix columns. So now the substitute bytes will be our substitution and shifting rows and mixing columns will be our permutations. And finally at the end of each round we are going to add our key.

When we have got 128 bits key, we have 10 of these rounds, when we have 192 bit key we have 12 of these rounds and when we have 256 bit key we have 14 of these rounds.

So that's just the overview of what AES does.

4. TOOL USED TO DEMONSTRATE AES ALGORITHM

4.1 PYTHON

Python is a general-purpose programming language. It is high level and easy to learn and dynamically initialized. It allows the programmers to write code in the object-oriented structural and the functional programming styles. It is a very concise programming language. Python supports adding codes of other languages into itself to run those functions which makes python more powerful i.e. embeddable and extensible. Python is interpreted line by line which means that the management of Memory, CPU and debugging of code much more simpler Like it is easy to interpret making debugging very easy. It can be easily integrated with other languages like C, C++, Java etc. It works on all the operating systems. It supports both Procedure-Oriented Programming and Object-Oriented Programming. Python is dynamically typed, and it also has automatic memory management. Memory Management in python involves a private heap containing all the python objects and the data structures. Python has interfaces to all the major databases. In the real world also if you face an issue into your code and have security for it such that access to them is restricted which helps you replicate the real-world scenarios. Python is so popular for its easy syntax and high-level features. The important feature of python is that it is an open-source language. It can be used to make anything like GUI application, Web application, Mobile application development, server site coding. Now with the rise of Artificial Intelligence & Machine Learning Algorithm python has come into the spotlight as it makes the work much more productive and easier.

How is Python interpreted? Python program is executed directly from the source code. The source code goes to an interpreter which turns it into an intermediate language first and then later into the native language or the machine language and then it gets executed.

Following Industries wherein python is implemented:

GOOGLE: Python provides better search results based on the ranking of the websites and much more.

DROPBOX: Dropbox is a cloud platform to save data online its client and server application are coded using python.

NETFLIX: It uses machine learning with python to learn the taste of its users cluster them into groups and then recommend them to show to retain them for longer.

NATIONAL SECURITY AGENCY: Cyber security analysis and other encryption and decryption work.

BITTORRENT: It has a peer-to-peer communication started out as a simple python file.

NASA: Scientist uses python to perform calculations which helps to reduce time.

4.2 We have used SPYDER using ANACONDA IDE as the version in python.

IDE stands for Integrated Development Environment. It is a software development tool. The work of IDE is to consolidate many functions together in a single framework, application or in a service. It also provides comprehensive facilities to develop your application.

SPYDER stands for Scientific Python Development Environment. The vital reason for electing Spyder IDE was that it is an open-source cross platform IDE. It is a very powerful editor. It is available through ANACONDA. Python Spyder IDE it comes as a default configuration along with the Anaconda Python distribution. It is a python distribution and an environment manager. In Spyder we can clear the variables automatically. SPYDER is a numerical computing environment. It is also used in providing tools for the data inspection it mainly highlights the syntax that provides the python console and many other. It has a unique combination of the advanced editing, analysis, debugging and profiling functionality. It is the best for the python IDE for windows operating system which allows us to run the python code by cell, line or file because of the interactive executions. Spyder also helps us to provide features like help, file explorer and find files. It also has an interactive way to trace every step of python code execution. Spyder provides us with automated code completion and also vertical and horizontal splitting.

ADVANTAGES:

- For programming tools like Spyder and Jupyter and for the importance of data science we use Anaconda as it is one stop destination.
- The command prompt of the Anaconda gives us the flexibility to install and use multiple libraries in the python.
- It is an open source.
- It is also used to support multiple popular data analytical language.
- Anaconda is very helpful as it is easy to deploy and implement the packages.
- It also provides better data visualization.
- It also has multiplatform support in simple words it helps in running on multiple operating systems.
- The containerization of the code is also very fast and easy.
- Anaconda is also helping us to visualize the workflows.

5. DATA ENCRYPTION AND DECRYPTION CODE USING SYMMETRIC AES ALGORITHM.

```
temp.py x AES.py* x
1  import json
2  from base64 import b64encode
3  from base64 import b64decode
4  from Crypto.Cipher import AES
5  from Crypto.Util.Padding import pad
6  from Crypto.Util.Padding import unpad
7  from Crypto.Random import get_random_bytes
8
9  bdata = input("Message : ")
10 data = bdata.encode('utf-8')
11 key = get_random_bytes(32)
12 print("key: ", key)
13 cipher = AES.new(key, AES.MODE_CBC)
14 ct_bytes = cipher.encrypt(pad(data, AES.block_size))
15 iv = b64encode(cipher.iv).decode('utf-8')
16 ct = b64encode(ct_bytes).decode('utf-8')
17 result = json.dumps({'iv':iv, 'ciphertext':ct})
18 print(result)
19
20 b64 = json.loads(result)
21 iv = b64decode(b64['iv'])
22 ct = b64decode(b64['ciphertext'])
23 cipher = AES.new(key, AES.MODE_CBC, iv)
24 bpt = unpad(cipher.decrypt(ct), AES.block_size)
25 pt = bpt.decode('utf-8')
26 print("The message was: ", pt)
```

Figure 5

OUTPUT 1

```
In [1]: runfile('C:/Users/mihee/Downloads/AES.py', wdir='C:/Users/mihee/Downloads')

Message : artificial intelligence
key:  b'\x8aa\xd7\x82\xed\xdf\xcbj\xfc5\xf5E\xdb1\xd7\x96\xcd3%\xff\xfa,\xb1\x98_
\x97\x82\xc0\x86\xb4\xcd\xb7'
{"iv": "abesL+xc+V8CIuuPgJsxhQ==", "ciphertext": "kP6nrQ85DGNngw4unky9Miex
+SC9jMsLKsBfhNYZHJ8="}
The message was:  artificial intelligence
```

Figure 6

OUTPUT 2

```
Message : artificial intelligence
key:  b'\x8aP\xa0Z7\xda\xb9\xe4\x19\x80\x99\x95\x89\xec\xee\x07\x95\xdc/\xd6\xee
\x8f\x81\xb6\x05\xa3\xeb\xc4\x99-\xda)'
{"iv": "C894SuVMqAhdgWPY20NzJw==", "ciphertext":
"L4zzqFy4iRYHWEk5v5YxUf0qFD9gxeEXKyZKNUFKfMc="}
The message was:  artificial intelligence
```

Figure 7

As we can see from Figure 6 and Figure 7, key and cipher text is changing for both the outputs, even though the plain text message is same.

6. CONCLUSION

So, from all the research we gathered, Artificial Intelligence is a powerful concept used in various Industries, especially in cyber security, as it encrypts the plain message into a cipher text which is nothing but a garble data consisting of random numbers, letters, and characters. We have demonstrated this concept by running AES algorithm in Python tool. We have observed that this Algorithm gives complex cipher text which is difficult to crack by the sniffer which in turn can be used to send the data over unsecured network.

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